# LUKE KURLANDSKI

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## RESEARCH SUMMARY

My Ph.D. research develops deep learning methods for malware detection and analysis, including language models trained on executable code, adversarial attacks on malware classifiers, and detectors that generalize across platforms. Going forward, I plan to continue research at the intersection of machine learning and security, with particular interest in the robustness of large language models and their applications to defensive security.

## EDUCATION

**Doctor of Philosophy (Ph.D.) @Rochester Institute of Technology (RIT)** *Aug. 2022 - May 2027*
- Department: Computing and Information Science
- Advised by: Dr. Matthew Wright & Dr. Yin Pan

**Bachelor of Science (B.S.) @The College of New Jersey (TCNJ)** *Aug. 2018 - May 2022*
- Department: Computer Science
- Minors: Mathematics, Physics

## PUBLICATIONS

### Under Review

- L. Kurlandski, D. Krisiloff, S. Coull & M. Wright, "Don't Just Eat the EXE: Raw-Binary Malware Classifiers Need Structure and Scale," Under review, 2026.
- C. Xie, L. Kurlandski, P. Miller & N. McLaughlin, "DAUGL: Cross-Platform Malware Detection via Unsupervised Global and Local Domain Alignment," Under review, 2026.

### Conference Proceedings

- L. Kurlandski, H. Berger, Y. Pan & M. Wright, "Beyond Raw Bytes: Towards Large Malware Language Models," *Network and Distributed System Security Symposium (NDSS)*, 2026.
- L. Kurlandski & M. Bloodgood, "Impact of Stop Sets on Stopping Active Learning for Text Classification," *IEEE International Conference on Semantic Computing (ICSC)*, 2022.
- D. Stolz, J. Strobelt, M. Leven, L. Kurlandski, H. Abourahma & D. McGee, "One-Step Fabrication of Surface Relief Dot-Matrix Holograms Using Supramolecular Azopolymer Thin Films," *SPIE Practical Holography: Displays, Materials, and Applications*, 2021.
- J. Strobelt, D. Stolz, M. Leven, L. Kurlandski & D. McGee, "Nanoimprint Lithography for the Replication of Optical Microstructures on Azopolymer Thin Films," *IEEE Conference on Lasers and Electro-Optics (CLEO)*, 2021.

### Journal Articles

- J. Strobelt, D. Stolz, M. Leven, M. Van Soelen, L. Kurlandski, H. Abourahma & D. McGee, "Optical Microstructure Fabrication Using Structured Polarized Illumination," *Optics Express*, 2022.

### Workshop Papers

- I. Ryan, L. Kurlandski & N. Mathews, "Towards Moving Target Defense for IoT Malware Detection," *Workshop on Software Engineering Research and Practices for the IoT (SERP4IoT)*, 2026.
- L. Kurlandski, R. Mosli, Y. Pan, S. Thianphan & M. Wright, "Do Fear the REAPIR: Adversarial Malware from API Replacement," *Workshop on Rethinking Malware Analysis (WoRMA)*, 2025.

### Poster Sessions

- L. Kurlandski, H. Berger, Y. Pan & M. Wright, "Pretrained Executable Language Models (ELMs) Excel at Malware Understanding," *USENIX Security Symposium (Poster)*, 2024.
- L. Kurlandski, N. Mathews, N. Hopper & M. Wright, "Modeling Stepping-Stone Flows Using Stochastic Random Walks and Black Box Neural Networks," *Western New York Image and Signal Processing Workshop (WNYISPW)*, 2024.
- L. Kurlandski, Y. Pan & M. Wright, "From Bytes to Insights: A New Era of Malware Detection," *Western New York Image and Signal Processing Workshop (WNYISPW)*, 2023.
- L. Kurlandski, Y. Pan & M. Wright, "An Adversarial Malware Evasion Attack Using Machine Translation," *AI@RIT Summit*, 2022.

- L. Kurlandski, "Evaluating ML Recommender Systems without Interactions Data," *ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2021.

## EXPERIENCE

**Machine Learning Security Group, RIT** *Aug. 2022 – May 2027*
*Graduate Research Assistant*; advised by Dr. Matthew Wright & Dr. Yin Pan.
- Researched adversarial examples, neural architecture design, and explainable machine learning for malware.

**Cybersecurity Division, CACI International** *Jun. 2021 – Aug. 2021*
*Research Intern*; advised by Anand Dinikar & Edward Phelps
- Developed deepfake audio detectors and contributed to forensic analysis of manipulated PDF documents.

**Learning Curve Group, TCNJ** *Feb. 2021 – May 2022*
*Undergraduate Researcher*; advised by Dr. Michael Bloodgood.
- Researched active learning and label-efficient machine learning for natural language processing.

**Collective Intelligence Group, TCNJ** *Jun. 2020 – Feb. 2021*
*Undergraduate Research Assistant*; advised by Dr. Monisha Pulimood.
- Built a recommendation system to help researchers find under-appreciated publications (NSF REU).

**Photonics Group, TCNJ** *Dec. 2018 – Feb. 2020*
*Undergraduate Researcher*; advised by Dr. David McGee.
- Wrote software to control lab hardware (lasers, motors, etc.) for optical microstructure fabrication.

## AWARDS & HONORS

**Provost's Outstanding Graduate Student Teaching Award, RIT** *Apr. 2025*
- Recognized primarily for teaching Advanced Malware Forensics, as well as TA work and mentoring.

**NSF Graduate Research Fellowship (GRFP)** *Apr. 2024*
- Three years of NSF support for graduate research in science and engineering.

**NDSEG Fellowship** *Mar. 2024*
- Three years of Department of Defense support for graduate research; declined in favor of the GRFP.

**AI Seed Fund Grant, RIT** *Jan. 2024*
- Wrote the $5,000 internal grant proposal that launched the research behind an NDSS 2026 paper.

**Goldberg-Neff Scholarship, TCNJ** *May 2022*
- Awarded to graduating seniors going on to graduate research.

**Phi Beta Kappa (honor society), TCNJ** *May 2022*
- Inducted into the national honor society for the liberal arts and sciences.

**Upsilon Pi Epsilon (honor society), TCNJ** *May 2022*
- Inducted into the international honor society for the computing disciplines.

**Department of Computer Science First-Year Award, TCNJ** *May 2019*
- Awarded by the computer science department to an outstanding first-year student.

**Merit Scholarship, TCNJ** *Aug. 2018 – May 2022*
- Four-year scholarship awarded to incoming students on academic merit.

**Dean's List, TCNJ** *Aug. 2018 – May 2022*
- Maintained a GPA above 3.5 in all eight undergraduate semesters.

## TEACHING & LEADERSHIP

**Instructor, RIT** *Jan. 2024 – May 2024*
- Taught a graduate research seminar on Advanced Malware Forensics as sole instructor; overhauled its materials.

**Graduate Teaching Assistant, RIT** *Aug. 2023 – Dec. 2023*
- Assisted with a graduate research seminar on Human Factors in Security and Privacy.

**Project Mentor, RIT** *Jan. 2023 – May 2027*
- Mentored 19 B.S. and M.S. students through 7 independent studies and 3 capstone projects.

**Founder, Kappa Delta Rho Nu Gamma Chapter, TCNJ** *Jan. 2019 – May 2022*
- Established the Nu Gamma Chapter of the Kappa Delta Rho fraternity at TCNJ.

**GED Instructor, Oasis – A Haven for Women and Children, Paterson, NJ** *Nov. 2017 – May 2018*
- Taught math, science, and English to adults preparing for the General Educational Development (GED) exam.

## SKILLS

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| **ML:** | PyTorch, Hugging Face (Transformers, Datasets), scikit-learn, TensorFlow, Captum |
| **Security:** | IDA Pro, Ghidra, radare2, OllyDbg, pefile, LIEF, Capstone |
| **Languages:** | Python, C, C++, Rust, Java, R, MATLAB, Mathematica |
| **Systems:** | Linux, SLURM / HPC clusters, Git, LaTeX |